

Chapter 4

Pricing of Exotic European Style Products

4.1 Exotic European Style Products

It will be interesting to see the impact of the different calibration methods studied in Chapter 3 on the prices of different derivative products. In this chapter we consider derivative instruments of European type in a slightly generalized sense. We speak of a European option when the cash-flows at pre-specified exercise dates (standard European product), or at a random date (stopping time) which is explicitly specified in the option contract.¹ Clearly, all European style products in this sense can be evaluated by straightforward Monte Carlo simulation. In the next subsections we introduce some popular exotic derivative products. We will consider all these products with respect to a Libor model in the spot Libor measure given by SDE (2.1). From Section 1.2.1 it is clear how to switch to a Libor SDE in some other numeraire measure.

4.1.1 Libor Trigger Swap

We consider the following version of a trigger swap over a period $[T_p, T_q]$ for a \$1 notional: Up to the first time after T_p , say T_τ , that the spot Libor $L_j(T_j)$ crosses a pre-specified trigger level K_j , the holder receives a spread rate s (or pays when s is negative) over $[T_p, T_\tau)$. Then, at T_τ the holder enters obligatorily into a swap contract with fixed rate κ for the remaining period

¹An American put option, for instance, does not fall into this class since an optimal stopping time depends on the model used for the underlying and thus cannot be specified in the contract.

$[T_\tau, T_q]$. Obviously, the value of the trigger swap at $t = 0$ is given by

$$Trswap(0) := E_{*}^{\mathcal{F}_0} \left(\sum_{j=p}^{\tau-1} \frac{s\delta_j}{B_*(T_{j+1})} + \frac{1}{B_*(T_\tau)} \sum_{j=\tau}^{q-1} (1 - B_q(T_\tau) - \kappa \sum_{j=\tau}^{q-1} \delta_j B_{j+1}(T_\tau)) \right), \quad (4.1)$$

in the spot Libor measure P^* (note that $B_*(0) = 1$). In expression (4.1), the spot Libor rolling over account as well as the zero bonds are evaluated exclusively at tenor times and therefore they can be expressed in the Libor process via the formulas

$$B_*(0) := 1, \quad B_*(T_i) = \frac{1}{B_1(0)} \prod_{p=1}^{i-1} (1 + \delta_p L_p(T_p)), \quad 1 \leq i \leq n, \quad (4.2)$$

$$B_j(T_i) = \prod_{p=i}^{j-1} (1 + \delta_p L_p(T_i))^{-1}, \quad 1 \leq i \leq j \leq n, \quad (4.3)$$

respectively (see (1.20) for the spot Libor process at an arbitrary time t). So the trigger swap value is estimated via Monte Carlo simulation of the processes $T_i \rightarrow B_*(T_i)$, $1 \leq i \leq n$, and $T_i \rightarrow B_j(T_i)$, $1 \leq i \leq j$, $j = 1, \dots, n$, via

$$\widehat{Trswap}_M(0) := \frac{1}{M} \sum_{m=1}^M \left(\sum_{j=p}^{\tau-1} \frac{s\delta_j}{B_*^{(m)}(T_{j+1})} + \frac{1}{B_*^{(m)}(T_\tau)} \sum_{j=\tau}^{q-1} (1 - B_q^{(m)}(T_\tau) - \kappa \sum_{j=\tau}^{q-1} \delta_j B_{j+1}^{(m)}(T_\tau)) \right). \quad (4.4)$$

In (4.4), the Monte Carlo sample of spot Libor trajectories $T_i \rightarrow B_*^{(m)}(T_i)$, $1 \leq i \leq n$, and zero bond trajectories $T_i \rightarrow B_j^{(m)}(T_i)$, $1 \leq i \leq j$, $j = 1, \dots, n$, $m = 1, \dots, M$, is obtained via (4.2) and (4.3), respectively, from a Monte Carlo simulation of the corresponding Libor process (2.1) in the spot Libor measure P^* .

4.1.2 Ratchet Cap

A ratchet cap with start tenor T_p and terminal tenor T_q is a standard cap over the period $[T_p, T_q]$, except that the strike κ_j for the period between T_j and T_{j+1} , which corresponds to the j -th caplet is the previous Libor plus a spread s , so $\kappa_j = L_{j-1}(T_{j-1}) + s$, for $p < j < q$. The first period strike is defined to be $\kappa_p = s + \kappa_0$, where κ_0 is some exogenously defined strike. The

value of the ratchet cap at $t = 0$ is thus given by

$$Rtcap(0) := E_{*}^{\mathcal{F}_0} \sum_{j=p}^{q-1} \frac{(L_j(T_j) - \kappa_j)^+ \delta_j}{B_*(T_{j+1})}, \quad (4.5)$$

in the spot Libor measure, where the spot numeraire B_* can be expressed in Libors by (4.2). So a Monte Carlo estimator for the ratchet cap is given by

$$\widehat{Rtcap}(0) := \frac{1}{M} \sum_{m=1}^M \sum_{j=p}^{q-1} (L_j^{(m)}(T_j) - \kappa_j)^+ \delta_j \frac{1}{B_*^{(m)}(T_{j+1})}, \quad (4.6)$$

where M is the number of Monte Carlo trajectories of the Libor process simulated from SDE (2.1).

4.1.3 Sticky Cap

A sticky cap with start tenor T_p and final tenor T_q is a standard cap over the period $[T_p, T_q]$, except that the strike κ_j for the period between T_j and T_{j+1} corresponding to the j -th caplet is the previous capped rate plus a spread s , so $\kappa_j = \min(L_{j-1}(T_{j-1}), \kappa_{j-1}) + s$, for $p < j < q$. As for the ratchet cap, the first period strike is defined to be $\kappa_p = s + \kappa_0$, for some exogenously defined strike κ_0 . The price of the sticky cap and the corresponding Monte Carlo estimator is then given by (4.5) and (4.6), respectively, again.

4.1.4 Auto-flex Cap

An auto-flex cap over a period $[T_p, T_q]$ with strike κ and pre-specified exercise number k can be regarded as a standard cap with strike κ over this period, except that only the first k In-The-Money caplets are paid off. The auto-flex cap price at $t = 0$ can thus be represented in the spot Libor measure by

$$Aflcap(0) := E_{*}^{\mathcal{F}_0} \sum_{j=p}^{q-1} 1_{\{\varkappa_j \leq k\}} (L_j(T_j) - \kappa)^+ \delta_j \frac{1}{B_*(T_{j+1})}, \quad (4.7)$$

where

$$\varkappa_j := \#\{i : L_i(T_i) > \kappa, \quad p \leq i \leq j\}, \quad p \leq j < q,$$

and the B_* values are expressed in Libors via (4.2). The corresponding Monte Carlo estimator is given by

$$\widehat{Aflcap}(0) := \frac{1}{M} \sum_{m=1}^M \sum_{j=p}^{q-1} 1_{\{\varkappa_j^{(m)} \leq k\}} (L_j^{(m)}(T_j) - K)^+ \delta_j \frac{1}{B_*^{(m)}(T_{j+1})}, \quad (4.8)$$

where M is the number of simulated Monte Carlo trajectories of the Libor process given by SDE (2.1).

4.1.5 Callable Reverse Floater

We now consider the callable reverse floater, an instrument which may behave depending on its particular specification like a floor (cap) or like a swaption (Jamshidian (1997)).

Let $\kappa, \kappa' > 0$. A *reverse floater* (RF) over a period $[T_p, T_q]$ is a contract for receiving $L_i(T_i)$ while paying $\max(\kappa - L_i(T_i), \kappa')$ at time T_{i+1} , for $i = p, \dots, q-1$, with respect to a unit principle, say \$1. A *callable reverse floater* (CRF) is an option to enter into a reverse floater at T_p . Hence, the CRF will be exercised at T_p when the value of the reverse floater at T_p is positive.

For the reverse floater, the T_{i+1} -cash-flows are thus given by

$$\begin{aligned} C_{T_{i+1}} &:= \delta_i L_i(T_i) - \delta_i \max(\kappa - L_i(T_i), \kappa') \\ &= \delta_i (L_i(T_i) - \kappa') - \delta_i \max(\kappa - \kappa' - L_i(T_i), 0). \end{aligned} \quad (4.9)$$

So, by (4.9), the reverse floater can be seen as a swap with strike κ' , *minus* a floor with strike $\kappa - \kappa'$. As a consequence, the value of the RF at time t , $t \leq T_p$ can be expressed as

$$\begin{aligned} RF(t) &= B_p(t) - B_q(t) - \kappa' \sum_{i=p}^{q-1} \delta_i B_{i+1}(t) \\ &\quad - \sum_{i=p}^{q-1} B_{i+1}(t) E_{i+1}^{\mathcal{F}_t} [\delta_i \max(\kappa - \kappa' - L_i(T_i), 0)] \end{aligned} \quad (4.10)$$

and therefore can be computed analytically in a Libor market model by Black (and Scholes) type expressions (1.31). The price of the callable reverse floater at time $t = 0$ is next given by

$$CRF(0) := E_*^{\mathcal{F}_0} \left[\frac{\max(RF(T_p), 0)}{B_*(T_p)} \right], \quad (4.11)$$

for which the corresponding Monte Carlo estimator is given by

$$\widehat{CRF}(0) := \frac{1}{M} \sum_{m=1}^M \frac{\max(RF^{(m)}(T_p), 0)}{B_*^{(m)}(T_p)},$$

for M Monte Carlo trajectories of the Libor process given by SDE (2.1). The zero bonds and spot numeraire involved in (4.11) can be expressed in Libors by (4.2) and (4.3), respectively.

Let us assume for simplicity that $\kappa' = 0$, so $C_{T_{i+1}} := \delta_i L_i(T_i) - \delta_i \max(\kappa - L_i(T_i), 0)$, and consider some special cases. When, for example, $\kappa \approx 2L_i(0)$, the probability that Libors exceed κ within the period $[0, T_q]$ can be neglected in practice, provided the time period $[0, T_q]$ is not too long. So, when the option is called the cash-flows are practically given by $C_{T_{i+1}} := 2\delta_i (L_i(T_i) - \kappa/2)$, and thus the option is basically a swaption on a doubled principal with

strike rate $\kappa/2$. If, in contrast, $\kappa \approx L_i(0)$ we may neglect the possibility that Libors fall below $\kappa/2$ and, practically speaking, the cash-flows $C_{T_{i+1}} := \delta_i L_i(T_i) - \delta_i \max(\kappa - L_i(T_i), 0)$ will be surely positive for every i . Then the option will be exercised in any case, yielding a cash-flow equal to the difference of the Libor rate on a forward loan with unit principle and the cash-flow of a floor over the period $[T_p, T_q]$ with strike κ . Therefore, in this situation the valuation of the CRF involves the valuation of a floor. We thus observe that the CRF has *both* cap/floor *and* swaption characteristics. Some more technical details around the callable reverse floater can be found in Schoenmakers & Coffey 1998. In Chapter 6 we consider the valuation of the callable reverse floater again in the context of lognormal Libor approximations.

4.2 Factor Dependence of Exotic Products

An important issue is the impact of a certain kind of Libor model calibration on the price of a particular exotic Libor product. More specifically, the question is: what is the effect of a particular type of calibration, for instance, a two, three, or full factor calibration to a given set of liquid instruments, on the price of a given exotic product. For the different calibration procedures presented in Chapter 3, we thus ask about the sensitivity of a Libor product value with respect to different ways of calibration to the same set of caps and swaptions. As a trivial observation, the members of the calibration set itself, for example the swaptions, are not sensitive as they are matched by the differently calibrated models. As pointed out by Andersen & Andreasen (2000a), also Bermudan swaptions, swaptions which are callable at a time to decide by the contract holder, and studied extensively in Chapter 5, are not very sensitive to the number of factors in the Libor model calibrated to the same set of swaptions. Below, however, we argue that the exotic cap structures introduced in Sections 4.1.2-4.1.4 are quite sensitive to a particular type of calibration. We give a detailed examination for the ratchet cap, from which it will be clear that the same effects apply to the other exotic cap structures. By experiments in Section 4.3 it turns out that the zero-adjusted Libor trigger swap is correlation/factor sensitive as well.

Correlation/factor dependence of the ratchet cap

Let us express the value of the ratchet cap at $t = 0$ in the form

$$\begin{aligned}
 Rtcap(0) &= \sum_{j=p}^{q-1} B_{j+1}(0) E_{j+1}^{\mathcal{F}_0} \delta_j (L_j(T_j) - \kappa_j)^+ \\
 &= \sum_{j=p}^{q-1} B_{j+1}(0) E_{j+1}^{\mathcal{F}_0} (\delta_j L_j(T_j) - \delta_j L_{j-1}(T_{j-1}) - \delta_j s)^+ \\
 &=: \sum_{j=p}^{q-1} B_{j+1}(0) E_{j+1}^{\mathcal{F}_0} (X_j - \delta_j s)^+, \tag{4.12}
 \end{aligned}$$

and concentrate on two consecutive forward Libors $L_{j-1}(T_{j-1})$ and $L_j(T_j)$ for a fixed j . We first write

$$\begin{aligned}
 X_j &= \delta_j L_j(T_j) - \delta_j L_{j-1}(T_{j-1}) = \delta_j L_j(T_j) \left(1 - \frac{L_{j-1}(T_{j-1})}{L_j(T_j)} \right) \\
 &\approx -\delta_j L_j(T_j) \ln \frac{L_{j-1}(T_{j-1})}{L_j(T_j)} \\
 &= \delta_j L_j(T_j) (\ln L_j(T_j) - \ln L_{j-1}(T_{j-1})), \tag{4.13}
 \end{aligned}$$

using $1 - z = -\ln z + O((z - 1)^2)$ for $z \rightarrow 1$. Next, we approximate the joint distribution of $Y_1 := \ln L_{j-1}(T_{j-1})$ and $Y_2 := \ln L_j(T_j)$ with respect to P_{j+1} by a two dimensional normal distribution,

$$(Y_1, Y_2) \stackrel{d}{\approx} (\mu_1 + \sigma_1(\varsigma_1 \sqrt{1 - \rho^2} + \varsigma_2 \rho), \mu_2 + \sigma_2 \varsigma_2), \tag{4.14}$$

where, in this analysis, μ_1 and μ_2 denote the mean, σ_1^2 and σ_2^2 the variances of Y_1 and Y_2 , respectively, ρ denotes the correlation between Y_1 and Y_2 , and ς_1, ς_2 are independent standard normal random variables. For a LMM we know that the distribution of Y_2 is exactly normal in P_{j+1} with $\mu_2 = 0$, and, that μ_1 is very small since in the drift of L_{j-1} there is only one term involved. For the present examination we therefore neglect μ_1 and then obtain by (4.13) and (4.14),

$$X_j \stackrel{d}{\approx} \delta_j e^{\sigma_2 \varsigma_2} \left((\sigma_2 - \rho \sigma_1) \varsigma_2 - \sigma_1 \sqrt{1 - \rho^2} \varsigma_1 \right). \tag{4.15}$$

We next compute and investigate the mean and variance of X_j , approximated by the r.h.s. of (4.15). Note that

$$E_{j+1}^{\mathcal{F}_0} X_j \approx E_{\varsigma_1, \varsigma_2} X_j = (\sigma_2 - \rho \sigma_1) \delta_j E_{\varsigma_2} \varsigma_2 e^{\sigma_2 \varsigma_2} = (\sigma_2^2 - \rho \sigma_1 \sigma_2) \delta_j e^{\sigma_2^2/2} \tag{4.16}$$

and that

$$\begin{aligned}
 E_{j+1}^{\mathcal{F}_0} X_j^2 &\approx E_{\varsigma_1, \varsigma_2} X_j^2 = E_{\varsigma_2} \delta_j^2 e^{2\sigma_2 \varsigma_2} E_{\varsigma_1} \left((\sigma_2 - \rho \sigma_1) \varsigma_2 - \sigma_1 \sqrt{1 - \rho^2} \varsigma_1 \right)^2 \\
 &= \delta_j^2 E_{\varsigma_2} e^{2\sigma_2 \varsigma_2} \left((\sigma_2 - \rho \sigma_1)^2 \varsigma_2^2 + \sigma_1^2 (1 - \rho^2) \right) \\
 &= \sigma_1^2 (1 - \rho^2) \delta_j^2 e^{2\sigma_2^2} + (\sigma_2 - \rho \sigma_1)^2 \delta_j^2 (1 + 4\sigma_2^2) e^{2\sigma_2^2},
 \end{aligned}$$

hence

$$\begin{aligned}\text{Var}(X_j) &:= \text{Var}_{\sigma_1, \sigma_2, \rho}(X_j)g \\ &= \sigma_1^2(1 - \rho^2)\delta_j^2 e^{2\sigma_2^2} + (\sigma_2 - \rho\sigma_1)^2 \delta_j^2 (1 + 4\sigma_2^2) e^{2\sigma_2^2} - (\sigma_2^2 - \rho\sigma_1\sigma_2)^2 \delta_j^2 e^{\sigma_2^2}.\end{aligned}\quad (4.17)$$

We then consider the derivative with respect to ρ ,

$$\begin{aligned}\frac{\partial}{\partial \rho} \text{Var}_{\sigma_1, \sigma_2, \rho}(X_j) \\ = \delta_j^2 e^{\sigma_2^2} \left(2\sigma_1\sigma_2^3 - 2\sigma_1\sigma_2 e^{\sigma_2^2} - 8\sigma_1\sigma_2^3 e^{\sigma_2^2} + 2\rho\sigma_1^2\sigma_2^2(4e^{\sigma_2^2} - 1) \right),\end{aligned}\quad (4.18)$$

which is linear in ρ with positive slope. Thus, (4.18) may be estimated from above by substituting $\rho = 1$, and then after some rearranging we obtain

$$\frac{\partial}{\partial \rho} \text{Var}_{\sigma_1, \sigma_2, \rho}(X_j) \leq -2\sigma_1\sigma_2\delta_j^2 e^{\sigma_2^2} \left(e^{\sigma_2^2} + \sigma_2(\sigma_2 - \sigma_1)(4e^{\sigma_2^2} - 1) \right).$$

Note that

$$\begin{aligned}e^{\sigma_2^2} + \sigma_2(\sigma_2 - \sigma_1)(4e^{\sigma_2^2} - 1) &\geq e^{\sigma_2^2} - \max(0, \sigma_1 - \sigma_2) \sigma_2(4e^{\sigma_2^2} - 1) \\ &\geq e^{\sigma_2^2} (1 - 4\sigma_2 \max(0, \sigma_1 - \sigma_2)) > 0\end{aligned}$$

when $4\sigma_2 \max(0, \sigma_1 - \sigma_2) < 1$. Hence,

$$\frac{\partial}{\partial \rho} \text{Var}_{\sigma_1, \sigma_2, \rho}(X_j) \leq -2\sigma_1\sigma_2\delta_j^2 e^{2\sigma_2^2} (1 - 4\sigma_2|\sigma_2 - \sigma_1|) < 0$$

for $4\sigma_2 \max(0, \sigma_1 - \sigma_2) < 1$, $\rho \leq 1$. As a result, when $4\sigma_2 \max(0, \sigma_1 - \sigma_2) < 1$ is satisfied, then $\text{Var}_{\sigma_1, \sigma_2, \rho}(X_j)$ decreases in ρ when $\rho \uparrow 1$.

In every Libor market model which is perfectly matched to caplets, we have $\sigma_2 = (\gamma_i^B)^2 T_i$, and $\sigma_1 \approx (\gamma_{i-1}^B)^2 T_{i-1}$ since $\mu_1 \approx 0$ (μ_1 deterministic would be enough for equality), where γ_{i-1}^B and γ_i^B are the respective caplet volatilities. Hence, σ_1 and σ_2 are not affected by a certain kind of calibration method discussed in Chapter 3. But, ρ is of course dependent on the correlation structure of the calibrated model.

In practice the condition $4\sigma_2 \max(0, \sigma_1 - \sigma_2) < 1$ comes down to

$$4\gamma_i^B \sqrt{T_i} \max(0, \gamma_{i-1}^B \sqrt{T_{i-1}} - \gamma_i^B \sqrt{T_i}) < 1. \quad (4.19)$$

Clearly, (4.19) is particularly fulfilled for those T_i where $(\gamma_i^B)^2 T_i \geq (\gamma_{i-1}^B)^2 T_{i-1}$, hence the T_i where the all-in caplet variance is increasing; see the discussion in Section 3.5.1. As we can see in Figure 3.3 this is in practice the case for small to mediate term T_i . Otherwise, when $\gamma_{i-1}^B \sqrt{T_{i-1}} = \gamma_i^B \sqrt{T_i}(1 + \epsilon)$, for some $\epsilon > 0$, condition (4.19) reads $4\epsilon(\gamma_i^B)^2 T_i < 1$. This roughly means that, if increasingness of the all-in caplet variance is violated, then the violation should not be too strong. Without further discussion we note that, typically, the latter situation applies, at least for the data sets considered in this book.

If we finally approximate X_j with a normal random variable $\tilde{X}$ with matching mean (4.16) and variance (4.17), we obtain for the j -th term in (4.12) the approximation

$$\begin{aligned} E_{j+1}^{\mathcal{F}_0} (X_j - \delta_j s)^+ &\approx E \left(\tilde{X} - \delta_j s \right)^+ \\ &= E \left((\sigma_2^2 - \rho \sigma_1 \sigma_2) \delta_j e^{\sigma_2^2/2} + \varsigma \sqrt{\text{Var}_{\sigma_1, \sigma_2, \rho}(X_j)} \right)^+, \end{aligned} \quad (4.20)$$

where ς is standard normally distributed. Now, for increasing ρ , the mean term of the Gaussian random variable between the brackets in the r.h.s. of (4.20) shifts into the negative direction, and in the meantime the variance decreases. From this it not difficult to show that the r.h.s. of (4.20) decreases with increasing ρ .

From the above heuristic analysis we conclude that, under some restrictions which are normally fulfilled in practice, the ratchet cap value decreases if the correlations between consecutive forward Libors increase. The latter typically happens when the number of factors in the calibration is decreased! Indeed, if one applies PCA analysis to a full rank correlation structure to reduce the number of factors to two or three, one typically observes that the instantaneous correlations at the first off-diagonal increase and tend to one. In the next section the factor sensitivity of the ratchet cap and other exotic caps is confirmed in different case studies.

REMARK 4.1 A typical effect of a two or three factor Principle Component Analysis applied to a full-rank correlation structure is the fact that correlations $\rho_{i,i+1}$, $\rho_{i,i+2}$, become very close to one while correlations $\rho_{i,i+k}$, for k large, are still comparable with the corresponding correlations of the original input matrix. So the effect of factor reduction should be distinguished from increasing ρ_∞ in a full rank correlation structure. $\square$

4.3 Case Studies

For typical examples of the products introduced in Sections 4.1.1–4.1.4 we here computed Monte Carlo prices with respect to differently calibrated Libor models. In particular, we consider for the data sets of 14.05.02, 03.06.02, 01.07.02, 08.08.02, the respective calibrations given in Tables 3.11–3.15, and price a set of exotic Libor products with respect to each calibration. For each calibrated data set we are going to price the following products by Monte Carlo simulation in the spot Libor measure.

- (i) *ATM Libor Cap*: Cap with maturity 1 yr., cap period 1-20 yr., strike equal to the (standard) forward swap rate over this period

(ii) *Auto Flex Cap*: As (i), but only the first 10 ITM caplets are paid off

(iii) *Sticky Cap*

- Case 1: Maturity 1 yr., sticky cap period 1-20 yr., exogenous start-up strike equal to the (standard) forward swap rate over this period, spread 20 bp.
- Case 2: Maturity 1 yr., sticky cap period 1-20 yr., exogenous start-up strike equal to the present Libor rate, spread 30 bp.

(iv) *Ratchet Cap*

- Case 1: Maturity 1 yr., sticky cap period 1-20 yr., exogenous start-up strike equal the present Libor rate, spread 30 bp.
- Case 2: As (iv)-Case 1, but with zero spread

(v) *Libor Trigger Swap* (knock-in)

- Case 1: Maturity 1 yr., period 1-20 yr., trigger level 6.5%, knock-in swap strike 6.5%, spread -30 bp. (where negative means to be paid)
- Case 2: As (v)-Case 1, with spread adjusted such that the contract value is zero

The results are presented in [Tables 4.2–4.9](#). In these tables each particular calibration case is denoted by one of the codes A1/A10–G1/G10, referring to a corresponding calibration case according to [Table 4.1](#).

Discussion of Results

We now discuss the pricing results given in [Tables 4.2–4.9](#), of the above mentioned example products with respect to different calibration methods, calibration instruments, and calibration dates. In this context we address the following issues.

- (a) Correlation/factor dependence of different products on a calibration type, with respect to a fixed set of calibration instruments at a fixed calibration date
- (b) Dependence of different product prices on the set of calibration instruments, for a fixed calibration method at a fixed calibration date
- (c) Dependence of different product prices on the calibration date, for a fixed calibration method and a fixed set of calibration instruments

Add. (a) For each calibration date and set of calibration instruments, 1 yr. or 1–10 yr. maturity swaptions, we conclude the following.

- The prices of the standard cap are consistent with each other, within the Monte Carlo error.
- The Auto-flex-cap coincides overall within the M.C. error on the full-factor calibration cases A1 and C1. For the cases A10 and C10 there is a coincidence within less than 3%, which is of the order of the typical RMS calibration fit for these calibrations. The 3-factor AFC prices due to B1 and B10 are about 5% below A1 and A10 prices, respectively. However, the prices due to calibrations with correlations constraint to $\rho_\infty > 0.5$, hence E1/10, G1/10, are 11-15% below the A1 and A10 prices. We thus conclude that the AFC is a (weakly) correlation sensitive product.
- For the sticky cap, the full-factor prices A1 and C1 differ slightly, 1-2%. The full-factor, LSq calibration with $\rho_\infty < 0.5$, D1,F1, give typically 5-7% below A1/C1. The E1,F1 calibrations ($\rho_\infty > 0.5$) give prices up to 17% below A1,C1. The 3-factor prices B1 fall about 17-20% below A1-C1 prices. For the corresponding 1-10 yr. calibrations, the discrepancies A10-C10 are a bit higher; the other effects are similar. Concluding, we may say that the sticky cap is strongly factor and correlation sensitive.
- The results for the ratchet cap confirm the correlation/factor analysis in Section 4.2. For the chosen examples the 3-factor calibration reduces the A/C full-factor prices even up to 25%. Further, similar conclusions apply as for the sticky cap. In particular, the ratchet cap is very correlation and factor sensitive.
- Also the trigger swap is sensitive for the number of factors and correlations. The relative sensitivities become extremely high of course when the TS is zero-adjusted.

Add. (b) Let us compare for fixed calendar date prices, due to calibrations A1-G1, hence 1 yr. maturity swaptions as calibration instruments, with prices due to A10-G10, 1-10 yr. maturity swaptions as calibration instruments. The caps almost coincide within the Monte Carlo error. The little difference must be caused by different systematic errors due to the SDE simulation. For the exotic caps we observe overall price differences in the order of 3-4% with an exception for the B columns (3-factor calibrations), where the differences are higher. An auxiliary test (not in the table) for trigger swap prices which are not zero-adjusted yielded similar results. Since the RMS calibration fit of A10-G10 is about 3-4% also, we may regard these prices as consistent with the corresponding A1-G1 values, except for the B column. However, since the A1-G1 calibrations are more accurate and the 1 yr. maturity calibration instruments are usually more liquid, the latter calibration procedures may be preferred.

Add. (c) By choosing the strikes of the exotic caps in dependence of the underlying yield curve for the different data sets, these products are more or

TABLE 4.1: Codes for calibration to different instruments at different calibration dates

Calibration Case	
A1/A10:	As in Table 3.14, 1 yr./10 yr. Column
B1/B10:	As in Table 3.15, 1 yr./10 yr. Column
C1/C10:	As in Table 3.13, 1 yr./10 yr. Column
D1/D10:	As in Table 3.11, 1 yr./10 yr. Column, $\rho < 0.5$
E1/E10:	As in Table 3.11, 1 yr./10 yr. Column, $\rho > 0.5$
F1/F10:	As in Table 3.12, 1 yr./10 yr. Column, $\rho < 0.5$
G1/G10:	As in Table 3.12, 1 yr./10 yr. Column, $\rho > 0.5$

TABLE 4.2: Exotic product prices for **X1**-calibrations, 14.05.02, base points (st. dev.)

Calibration	→	14		05	02	
		A1	B1	C1	D1	E1
Cap		921 (8)	925 (9)	911 (8)	914 (8)	920 (9)
Auto Flex Cap		444 (3)	425 (3)	440 (3)	436 (3)	399 (3)
Sticky Cap 1		1074 (6)	888 (6)	1078 (6)	1050 (6)	939 (6)
Sticky Cap 2		985 (5)	815 (5)	995 (5)	956 (5)	865 (5)
Ratchet Cap 1		377 (3)	282 (2)	373 (3)	359 (3)	323 (3)
Ratchet Cap 2		531 (3)	434 (2)	537 (3)	517 (3)	473 (3)
Trigger Swap		3 (7)	91 (7)	3 (7)	16 (7)	131 (5)
Trsw. Spread		-30	-30	-30	-30	-30
Zero-Adj. Spread		-30	-50	-30	-35	-60

less standardised. We therefore observe a smooth behaviour of (exotic) cap prices, when going from one datum to the next.

TABLE 4.3: Exotic product prices for **X10**-calibrations, 14.05.02, base points (st. dev.) (one standard deviation)

	14		05	02	
Calibration →	A10	B10	C10	D10	E10
Cap	908 (8)	905 (8)	901 (8)	920 (9)	908 (9)
Auto Flex Cap	460 (3)	436 (3)	446 (3)	421 (3)	399 (3)
Sticky Cap 1	1125 (6)	958 (6)	1097 (5)	1023 (6)	946 (6)
Sticky Cap 2	1034 (5)	866 (5)	998 (5)	938 (5)	893 (6)
Rachet Cap 1	405 (3)	317 (3)	387 (3)	353 (3)	336 (3)
Rachet Cap 2	551(3)	463 (2)	541 (3)	513 (3)	487 (3)
Trigger Swap	-23 (7)	54 (7)	-30 (7)	56 (6)	107 (5)
Trsw. Spread	-30	-30	-30	-30	-30
Zero-Adj. Spread	-22	-45	-25	-44	-55

TABLE 4.4: Exotic product prices for **X1**-calibrations, 03.06.02, base points (st. dev.)

	03	06	02		
Calibration →	A1	B1	C1	D1	E1
Cap	928 (8)	916 (9)	931 (8)	941 (8)	925 (9)
Auto Flex Cap	447 (3)	426 (3)	441 (3)	431 (3)	407 (3)
Sticky Cap 1	1138 (6)	901 (6)	1106 (6)	1058 (6)	971 (6)
Sticky Cap 2	1001 (5)	806 (5)	983 (5)	956 (5)	874 (5)
Rachet Cap 1	382 (3)	287 (2)	378 (3)	366 (3)	332 (3)
Rachet Cap 2	538 (3)	442 (2)	535 (3)	520 (3)	485 (3)
Trigger Swap	1 (8)	98 (7)	2 (7)	26 (7)	101 (6)
Trsw. Spread	-30	-30	-30	-30	-30
Zero Adj. Spread	-30	-50	-30	-36	-55

TABLE 4.5: Exotic product prices for **X10**-calibrations, 03.06.02, base points (st. dev.)

Calibration	→	03		06	02	
		A10	B10	C10	D10	E10
Cap		924 (8)	924 (8)	916 (8)	928 (8)	930 (9)
Auto Flex Cap		461 (3)	437 (3)	456 (3)	428 (3)	397 (3)
Sticky Cap 1		1149 (6)	960 (6)	1114 (5)	1045 (6)	952 (6)
Sticky Cap 2		1026 (5)	860 (5)	1000 (5)	960 (5)	877 (5)
Ratchet Cap 1		393 (3)	319 (3)	385 (3)	369 (3)	336 (3)
Ratchet Cap 2		555 (3)	473 (3)	539 (3)	527 (3)	488 (3)
Trigger Swap		30 (7)	65 (7)	-16 (7)	44 (6)	115 (6)
Trsw. Spread		-30	-30	-30	-30	-30
Zero-Adj. Spread		-25	-40	-25	-42	-55

TABLE 4.6: Exotic product prices for **X1**-calibrations, 01.07.02, base points (st. dev.)

Calibration	→	01		07	02	
		A1	B1	C1	F1	G1
Cap		958 (9)	963 (10)	969 (9)	958 (9)	981 (10)
Auto Flex Cap		459 (4)	442 (4)	465 (4)	442 (4)	422 (4)
Sticky Cap 1		1089 (6)	929 (7)	1113 (6)	1022 (6)	969 (6)
Sticky Cap 2		936 (6)	786 (6)	956 (6)	901 (6)	857 (6)
Ratchet Cap 1		353 (3)	249 (2)	353 (3)	326 (3)	298 (3)
Ratchet Cap 2		510 (3)	253 (2)	513 (3)	486 (3)	462 (3)
Trigger Swap		-48 (7)	49 (7)	-45 (8)	-2 (7)	49 (6)
Trsw. Spread		-30	-30	-30	-30	-30
Zero-Adj. Spread		-20	-40	-20	-30	-40

TABLE 4.7: Exotic product prices for **X10**-calibrations, 01.07.02, base points (st. dev.)

Calibration	→	01		07	02	
		A10	B10	C10	F10	G10
Cap		959 (9)	961 (9)	966 (9)	957 (10)	977 (10)
Auto Flex Cap		471 (4)	451 (4)	475 (4)	423 (3)	420 (4)
Sticky Cap 1		1140 (6)	970 (6)	1155 (6)	1040 (6)	1006 (6)
Sticky Cap 2		993 (5)	843 (6)	990 (5)	898 (6)	865 (6)
Ratchet Cap 1		387 (3)	291 (3)	389 (3)	329 (3)	325 (3)
Ratchet Cap 2		538 (3)	445 (3)	542 (3)	483 (3)	480 (3)
Trigger Swap		-103 (8)	3 (7)	-96 (8)	28 (6)	44 (6)
Trsw. Spread		-30	-30	-30	-30	-30
Zero-Adj. Spread		-10	-30	-10	-35	-37

TABLE 4.8: Exotic product prices for **X1**-calibrations, 08.08.02, base points (st. dev.)

Calibration	→	08		08	02	
		A1	B1	C1	F1	G1
Cap		995 (10)	977 (10)	981 (10)	991 (10)	997 (10)
Auto Flex Cap		447 (4)	426 (4)	443 (4)	430 (4)	413 (4)
Sticky Cap 1		1053 (7)	936 (7)	1067 (6)	989 (7)	963 (7)
Sticky Cap 2		856 (6)	716 (6)	847 (6)	783 (6)	757 (6)
Ratchet Cap 1		307 (3)	227 (2)	306 (3)	277 (3)	266 (3)
Ratchet Cap 2		464 (3)	384 (3)	467 (3)	437 (3)	424 (3)
Trigger Swap		-45 (7)	37 (7)	-47 (7)	20 (6)	53 (6)
Trsw. Spread		-30	-30	-30	-30	-30
Zero-Adj. Spread		-25	-35	-23	-32	-38

TABLE 4.9: Exotic product prices for **X10**-calibrations, 08.08.02, base points (st. dev.)

Calibration	→	08		08	02	G10
		A10	B10	C10	F10	
Cap		997 (9)	986 (10)	1000 (9)	989 (10)	1010 (10)
Auto Flex Cap		453 (4)	446 (4)	456 (4)	433 (4)	413 (4)
Sticky Cap 1		1153 (7)	1001 (7)	1120 (7)	1073 (7)	1019 (7)
Sticky Cap 2		923 (6)	797 (6)	901 (6)	872 (6)	815 (6)
Ratchet Cap 1		346 (3)	273 (3)	338 (3)	318 (3)	309 (3)
Ratchet Cap 2		501 (3)	434 (3)	494 (3)	482 (3)	478 (3)
Trigger Swap		-106 (7)	-25 (7)	-96 (7)	-30 (6)	12 (5)
Trsw. Spread		-30	-30	-30	-30	-30
Zero-Adj. Spread		-10	-27	-12	-26	-32

